Book VI

Effective Continuum and Calculus Legitimacy

Elements of Nested Fibrational Cosmology

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VI.0 Purpose

Book VI is the interface legitimacy book of the canonical spine.

Books I–V have established: the observational quotient (Book I), local boundary law (Book II), transport structure (Book III), the universal source **U** (Book IV), and the law of lawful descent to branches (Book V). None of these books licenses continuum, differential, or integral language. That language must be earned; it cannot be smuggled in.

The central question Book VI answers is: *when* is it legitimate for a branch book to write differential equations, action functionals, integral operators, smooth fields, or geometric structures? The answer is: only when the relevant certified observable families are asymptotically coherent in the sense made precise here, only on the certified observable class, and only within a declared continuum interface regime.

Book VI establishes this legitimacy through a three-stage progression. First, it defines certified observable families and the scale-normalized increments that are their discrete analogues of derivatives. Second, it proves that asymptotically coherent observable families admit effective smooth representatives. Third, it establishes when differential and integral notation are legitimate as shorthand for the asymptotic behavior of the underlying discrete structure.

The governing theorem of the book is the Effective Continuum Legitimacy Theorem: differential and integral notation are lawful if and only if they faithfully encode the asymptotic behavior of scale-normalized increments and accumulators within a declared continuum interface regime.

Book VI is a tool-legitimacy book. It does not prove closure of any specific endpoint program. It does not identify which observable families satisfy the interface hypotheses — that burden remains partly branch-dependent. It licenses the tools; the branches must verify eligibility.

1. Imports and Standing Rules

Imports. All certified results of Books I–V. In particular: the observational quotient and defect bookkeeping (Book I), the boundary law and capacity structure (Book II), the Unified Coercive-Transport Inequality and observational-transfer objects (Book III), the POT category and universal source **U** (Book IV), and the branch legitimacy law (Book V).

Standing Rule 1.1 ([D]). Continuum language is licensed only as an effective encoding of certified discrete structure. It is not primitive.

Standing Rule 1.2 ([D]). Differential notation may not appear in a canonical branch text unless the relevant scale-normalized increment structure has been certified in the declared interface regime.

Standing Rule 1.3 ([D]). Integral notation may not appear unless the relevant certified accumulation or transport structure has been established.

Standing Rule 1.4 ([D]). Continuum language is valid only on the certified observable class and only within the declared continuum interface regime.

Standing Rule 1.5 ([D]). Refinement suggestiveness is not enough; asymptotic coherence or effective convergence must be proved at the required level before continuum language acquires canonical force.

2. Certified Observable Families and Discrete Structure

Definition 2.1 ([D]). A *certified observable family* $\mathcal{F} = \{f_R\}_{R\in\mathrm{Reg}}$ is a collection of observables, one for each admissible regime R, such that: each $f_R \in O_R$ is a realized stabilized local observable outcome; and the family is transfer-compatible in the sense of Book III, Definition III.2.6.

Lemma 2.2 ([U]). *Certified observable families are invariant under replacement by quotient-equivalent and branch-equivalent representatives.*

Proof. Each f_R is a stabilized local observable outcome, hence a function on the stabilized quotient Σ_∞ (Book I, Theorem I.6.1). Quotient-equivalent representatives are in the same class; therefore f_R assigns them the same value. Branch-equivalence is preserved by the transfer compatibility condition of the family. □

Definition 2.3 ([D]). A *scale-normalized increment* of a certified observable family $\mathcal{F}$ at stage n on a region U is

$$h_n(U) \;:=\; \frac{f_{R_{n+1}}(U) - f_{R_n}(U)}{\ell_n},$$

where $\ell_n > 0$ is the declared comparison length at stage n. The scale-normalized increment measures the change in the observable per unit comparison length under one admissible enlargement step.

Lemma 2.4 ([U]). *Scale-normalized increments, when defined, depend only on certified comparison data.*

Proof. The numerator $f_{R_{n+1}}(U) - f_{R_n}(U)$ involves only realized stabilized local outcomes, which are quotient-visible by Lemma 2.2. The denominator ℓ_n is a declared comparison length, hence part of the certified data of the interface regime. The quotient depends only on these certified quantities. □

Definition 2.5 ([D]). The *discrete gradient datum* $d_n(U)$ of a certified observable family on a region U at stage n is the vector of scale-normalized increments in all declared comparison directions.

Definition 2.6 ([D]). The *discrete flux datum* $J_n(U)$ on a region U at stage n is the certified flow of observable content across the collar $\partial_{\rho^*}U$, determined by the LS-2 boundary-law structure of Book II.

Definition 2.7 ([D]). A *discrete accumulator* $A_n(U)$ on a region U is the finite sum

$$A_n(U) \;:=\; \sum_{k=0}^{n} J_k(U)\cdot \ell_k,$$

formed by accumulating flux data with declared comparison lengths. It is the discrete analogue of a line or volume integral.

LEMMA 2.8 ([U]). *Discrete gradient data and discrete flux data are presentation-independent once defined on a certified observable family.*

PROOF. Both are defined in terms of stabilized local observable outcomes and the LS-2 boundary law, all of which are quotient-visible. Therefore they are independent of the particular presentation representative chosen within each equivalence class. □

LEMMA 2.9 ([U]). *Discrete accumulators formed from certified flux data are invariant under lawful decomposition refinement.*

PROOF. Decomposition refinement is an admissible enlargement. Under lawful admissible enlargement, the LS-2 boundary law (Book II, Theorem II.4.1) and the defect-ledger additivity (Book I, Proposition I.5.4) together ensure that the flux data and accumulated totals are preserved: summing over a refined decomposition gives the same total as summing over the coarser one. □

DEFINITION 2.10 ([D]). A certified observable family is *refinement-consistent* if the scale-normalized increments $h_n(U)$ stabilize under admissible refinement: there exists a stage N beyond which $h_n(U)$ changes by less than any declared tolerance $\varepsilon > 0$ at each further admissible refinement step.

LEMMA 2.11 ([U]). *Refinement consistency of a certified observable family is preserved under lawful source-descended comparison maps.*

PROOF. Under a lawful source-descended comparison map (a transfer map in the sense of Book III), the certified observable content is preserved up to the declared transfer structure. Therefore if the increments stabilize in the source regime, the transferred increments stabilize in the target regime by the transfer-compatibility conditions of Book III, Definition III.2.6. □

3. Asymptotic Coherence and Effective Smooth Representatives

DEFINITION 3.1 ([D]). A certified observable family $\mathcal{F}$ is *asymptotically coherent* if the scale-normalized increments $h_n(U)$ form a Cauchy sequence in the declared comparison norm as $n \to \infty$: for every declared tolerance $\varepsilon > 0$, there exists N such that $|h_m(U) - h_n(U)| < \varepsilon$ for all $m, n \geq N$.

DEFINITION 3.2 ([D]). An *effective smooth representative* of an asymptotically coherent certified observable family $\mathcal{F}$ is a smooth function $\phi : \mathcal{R} \to \mathbb{R}$ defined on a declared interface regime $\mathcal{R}$ such that:

(1) the values $\phi(x)$ approximate the certified observable data f_{R_n} within the declared tolerance at each stage n; and
(2) the differential expressions of ϕ approximate the scale-normalized increments h_n in the declared comparison norm.

The effective smooth representative encodes the observable data; it does not replace or displace it.

DEFINITION 3.3 ([D]). A *continuum interface regime* is a declared pair $(\mathcal{R}, \mathcal{F})$ consisting of a smooth domain $\mathcal{R}$ and a certified observable family $\mathcal{F}$ such that:

(1) $\mathcal{F}$ is asymptotically coherent on $\mathcal{R}$;
(2) the scale-normalized increments converge to the relevant differential expressions of the effective smooth representative on $\mathcal{R}$; and
(3) the discrete accumulators converge to the corresponding integral expressions on $\mathcal{R}$.

LEMMA 3.4 ([U]). *If asymptotic coherence fails, then no effective smooth representative is canonically licensed.*

PROOF. An effective smooth representative requires (Definition 3.2) that differential expressions approximate scale-normalized increments. If asymptotic coherence fails, the scale-normalized increments do not converge; therefore no smooth function's derivatives can approximate them, and no effective smooth representative exists in the declared regime. □

THEOREM 3.5 ([C]). (Refinement Coherence Theorem.) *A certified observable family satisfying the declared normalization and persistence conditions is asymptotically coherent under admissible refinement.*

PROOF. By Definition 3.1, asymptotic coherence requires the scale-normalized increments to form a Cauchy sequence. Under the declared normalization conditions (specifically, that $\ell_n \to 0$ at a controlled rate relative to the refinement steps) and the persistence conditions (that the transfer-compatibility of Lemma 2.11 holds), the increments $h_n(U)$ form a bounded sequence that changes by less than ε per step once sufficiently refined. This Cauchy property follows from the Finite Stabilization Theorem (Book I, Theorem I.6.1) applied to the quotient of increment values: the quotient stabilizes, so the increments eventually lie in a common equivalence class, forcing convergence. □

THEOREM 3.6 ([C]). (Effective Representative Existence Theorem.) *If a certified observable family is asymptotically coherent and satisfies the declared compactness or stabilization conditions on the continuum interface regime, then it admits an effective smooth representative.*

PROOF. By asymptotic coherence (Definition 3.1), the scale-normalized increments converge in the declared norm. The declared compactness or stabilization conditions ensure that this convergence is uniform on the interface regime. By a standard completion argument in the relevant function space, the limiting object is a smooth function ϕ satisfying the approximation conditions of Definition 3.2. □

THEOREM 3.7 ([C]). (Effective Representative Uniqueness Theorem.) *Any two effective smooth representatives of the same certified observable family are equivalent with respect to the preserved asymptotic content.*

PROOF. Suppose ϕ and ϕ' are two effective smooth representatives of the same certified observable family on the same interface regime. By Definition 3.2, both ϕ and ϕ' have their differential expressions converging to the same scale-normalized increments $h_n(U)$. Therefore ϕ and ϕ' have equal derivatives in the limit, and since both are smooth on a connected domain, they agree up to a constant. The constant is fixed by

the constraint that both approximate the same certified observable values f_{R_n}. Hence $\phi = \phi'$. □

4. Differential and Integral Legitimacy

PROPOSITION 4.1 ([C]). *Whenever an effective smooth representative ϕ exists, its admissible differential expressions must be interpretable as shorthand for certified scale-normalized increment behavior.*

PROOF. By Definition 3.2(2), the differential expressions of ϕ approximate the scale-normalized increments h_n. Therefore any differential expression of ϕ is, to within the declared tolerance, a re-encoding of h_n. It cannot carry more observational content than the underlying scale-normalized increment. □

THEOREM 4.2 ([C]). (Effective Differential Legitimacy Theorem.) *For a certified observable family in a lawful continuum interface regime, differential notation is legitimate if and only if it faithfully encodes the asymptotic behavior of scale-normalized increments.*

PROOF. *Sufficiency:* If differential notation faithfully encodes the asymptotic behavior of scale-normalized increments, then by Theorem 3.6 an effective smooth representative exists, and its differential expressions are the legitimate continuum encoding of the discrete increment data.

Necessity: If differential notation does not faithfully encode the asymptotic behavior of scale-normalized increments, then the smooth function whose derivatives are that notation is not an effective smooth representative (it fails condition (2) of Definition 3.2). Therefore the differential notation lacks the asymptotic foundation required for legitimacy. □

COROLLARY 4.3 ([C]). *No derivative claim has canonical force outside the certified observable class and declared interface regime.*

COROLLARY 4.4 ([C]). *Any downstream PDE expression has theorem-level canonical force only to the extent that its differential terms are backed by certified normalized increment structure.*

REMARK 4.5 ([R]). Corollary 4.4 is the standard citation for branch papers that use PDE language. When a branch book writes a partial differential equation — whether Navier–Stokes, Yang–Mills, Einstein field equations, or any other — it must cite the continuum interface regime in which that equation's differential terms are backed by certified discrete increment structure. Without such a citation, the PDE notation has no canonical force within the NFC program.

PROPOSITION 4.6 ([C]). *Whenever an effective smooth representative ϕ exists, its admissible integral expressions must be interpretable as shorthand for certified discrete accumulation or transport structure.*

PROOF. By Definition 3.2(1), the values of ϕ approximate the certified observable data at each stage. By Lemma 2.9, discrete accumulators $A_n(U)$ are refinement-invariant. Integral expressions of ϕ therefore approximate $A_n(U)$ and re-encode discrete

accumulation data without carrying more observational content than the underlying discrete structure. □

THEOREM 4.7 ([C]). (Effective Integral Legitimacy Theorem.) *For a certified observable family in a lawful continuum interface regime, integral notation is legitimate if and only if it faithfully encodes certified accumulation or transport data.*

PROOF. Parallel to the proof of Theorem 4.2, replacing scale-normalized increments with discrete accumulators and derivatives with integrals throughout. □

COROLLARY 4.8 ([C]). *No integral identity has canonical force outside the certified observable class and declared interface regime.*

COROLLARY 4.9 ([C]). *Variational or conservation-style integral statements are lawful only insofar as they descend from certified discrete accumulation structure. This governs all action principles and balance laws in branch books.*

5. The Continuum Bridge Schema

The previous sections establish when continuum notation is legitimate for a single certified observable family. Branch books typically need to *promote* a result from the certified discrete regime to a continuum statement. This promotion requires an explicit bridge schema.

DEFINITION 5.1 ([D]). A *scoped certificate* for continuum promotion is a certified result whose force is limited to the explicit regime named in its statement: the specific certified observable family, the specific interface regime, and the specific tolerance bounds declared at the time of the result.

DEFINITION 5.2 ([D]). A *continuum bridge* is a separately proved theorem carrying a scoped certificate from a declared discrete or pre-continuum regime into a target continuum regime, with explicit specification of:

(1) the source-side certified discrete burden,
(2) the target continuum criterion being promoted to, and
(3) the scope of the promoted claim.

THEOREM 5.3 ([U]). (Continuum Bridge Schema Theorem.) *A branch may promote a certified discrete result to continuum force only when:*

(1) *a scoped certificate establishes the result on the source discrete regime, and*
(2) *a separately proved continuum bridge carries that certified content to the declared continuum target.*

Neither ingredient alone suffices. A scoped certificate alone establishes only the discrete claim; a continuum bridge alone has no certified source content to carry.

PROOF. A scoped certificate proves only the claim on its declared discrete regime; it carries no automatic asymptotic or continuum extension (Standing Rule 1.5). A continuum bridge, by Definition 5.2, must be "separately proved" — it cannot be the same proof as the scoped certificate. Therefore both are needed. The combination provides the source certification and the bridge, jointly licensing the promoted claim. □

Corollary 5.4 ([U]). *Finite certification does not self-promote to continuum force.*

Corollary 5.5 ([U]). *Source-side canonical structure does not self-promote to physical endpoint identification.*

Corollary 5.6 ([U]). *Branch-conditional bridge success does not self-promote to unconditional continuum closure.*

Theorem 5.7 ([U]). (Continuum No-Smuggling Corollary.) *Any use of differential, integral, field, manifold, or variational language beyond the scope of a certified continuum interface regime is non-canonical and has no theorem-level force in the NFC program.*

Proof. By Theorem 4.2, differential notation is legitimate only within a certified interface regime. By Theorem 4.7, integral notation is legitimate only within a certified interface regime. By Theorem 5.3, promotion from discrete to continuum requires both a scoped certificate and a continuum bridge. Any use of such notation outside these conditions lacks the legitimacy foundation and therefore has no theorem-level force. □

Corollary 5.8 ([C]). *A downstream branch may use suggestive continuum language before full interface certification, but such language must be explicitly marked as heuristic. Heuristic use does not constitute a canonical claim.*

Proof. Theorem 5.7 prohibits canonical use of continuum language outside certified regimes. However, it does not prohibit exploratory or motivational language, provided such language is explicitly labeled as heuristic and not cited as having theorem-level force. The marking requirement follows from Book VII's status discipline, which prohibits silent status inflation. □

6. Governing Theorem of Book VI

Theorem 6.1 ([C]). (Effective Continuum Legitimacy Theorem.) *For certified stabilized observables in a lawful continuum interface regime, differential and integral notation are lawful if and only if they faithfully encode the asymptotic behavior of scale-normalized increments and certified accumulators, respectively, within that declared regime.*

Proof. The differential direction is Theorem 4.2. The integral direction is Theorem 4.7. Together they give the full biconditional over both differential and integral language. □

Corollary 6.2 ([C]). *For a certified observable family in a lawful continuum interface regime, the full calculus of differential and integral operators is jointly legitimate as asymptotic encoding of the certified discrete dynamics.*

Remark 6.3 ([R]). (E-CONT bridge obligations and discrete Sobolev surrogate.) *(Pre-canonical remarks have no canonical force; see Book VII, Standing Rule* **??***.)* The pre-canonical program provides two confirmations that Book VI's Continuum Bridge Schema (Theorem 5.3) correctly gates the YM and gravity continuum programs:

(i) **E-CONT bridge obligations** (v7.35 §§11051, 11067): the encoding-map bridge ($\Gamma : \mathcal{K}(G_n) \to M$) and curvature-proxy bridge (E3-proxy) that appear in the YM continuum program (R3.6) and the gravity continuum program (R1.1–R1.2) are both instances of the Continuum Bridge Schema. Book VI's `thm:continuum-bridge-schema` is therefore the canonical formulation that both programs must cite.

(ii) **NS-4B Sobolev embedding surrogate** (v7.35 §9333): The inequality $\|f\|_\infty \leq C_S(s)\|f\|_{H^s}$ with table-computable constant $C_S(s) = F_S(s, d_{\text{eff}}, C_{\text{HK}})$ is derived from the ACA Route B → Hypercontractivity → Nash chain from finite NF_2 data alone. It is the discrete-native surrogate for the Sobolev embedding used in the NS and YM continuum arguments, and belongs under Book VI's licensed continuum bridge infrastructure.

Both items are confirmed instances of existing Book VI machinery; they require no new canonical material.

7. Boundary of Book VI

What Book VI proves.

(1) Scale-normalized increments and discrete accumulators are well-defined, POT-native, and presentation-independent.
(2) Asymptotically coherent certified observable families admit effective smooth representatives, unique up to preservation of asymptotic content.
(3) Differential notation is legitimate iff it faithfully encodes scale-normalized increment asymptotics within a declared interface regime.
(4) Integral notation is legitimate iff it faithfully encodes certified accumulator asymptotics within a declared interface regime.
(5) The Continuum Bridge Schema governs all promotion from discrete to continuum force.
(6) No continuum language is canonical outside a declared and certified interface regime.

What Book VI does not prove, and may not be assumed here.

(1) *Which observable families satisfy the interface hypotheses.* The burden of verifying asymptotic coherence and the compactness or stabilization conditions is branch-specific. Branch books must establish these conditions for their own observable families before citing the Effective Continuum Legitimacy Theorem.
(2) *Branch closures.* Licensing continuum language does not constitute closing any specific branch endpoint program. Yang–Mills, Navier–Stokes, and gravity closure still require branch-specific bridge theorems.
(3) *Primitive ontology.* The existence of an effective smooth representative is never proof that the continuum is ontologically primitive. The continuum is an effective encoding, not a foundation.
(4) *Action principles, gauge theories, or field equations by default.* No specific PDE, Lagrangian, or field equation is licensed by this book. Each requires its own declared interface regime and its own bridge theorem.

8. Handoff to Book VII

Book VI establishes when continuum tools are licensed. Book VII establishes how results using those tools may be cited, scoped, and combined. The critical connection is that Book VII's promotion law governs the same promotion steps that Book VI's Continuum Bridge Schema describes structurally: Book VI tells you when the bridge is possible; Book VII tells you how to state and cite the result once it has been built.

Every branch book that uses results from Book VI must also conform to Book VII's closure schema. The two books work in tandem: interface legitimacy and closure discipline are complementary, not alternative, requirements.

9. Dependency Ledger

Theorem	St.	Depends on	Exports to
2.2 Cert. fam. quotient-inv.	[U]	Bk. I Thm., Bk. III Def.	2.4, 6.1
2.4 Increments cert.-data	[U]	2.2, Def. 2.3	4.2, 6.1
2.8 Flux pres.-indep.	[U]	Bk. II LS-2, Bk. I quotient	2.9, 4.7
2.9 Accumulator inv.	[U]	2.8, Bk. I ledger add.	4.7, 6.1
3.4 Incoherence blocks smooth rep.	[U]	Defs. 3.1, 3.2	3.6
3.5 Refinement Coherence Thm.	[C]	Defs. 2.1–3.1	3.6, 6.1
3.6 Eff. Rep. Existence Thm.	[C]	3.5, compactness	3.7, 4.2, 4.7
3.7 Eff. Rep. Uniqueness Thm.	[C]	3.6	6.1
4.2 Diff. Legitimacy Thm.	[C]	3.6, SR 1.2	4.4, 6.1
4.7 Int. Legitimacy Thm.	[C]	2.9, 3.6	4.9, 6.1
5.3 Continuum Bridge Schema Thm.	[U]	Def. 5.2, SR 1.5	5.7, 5.4; Bk. VII; all branch books
5.7 Continuum No-Smuggling Corollary	[U]	4.2, 4.7, 5.3	Constitutional rule for all branch books
6.1 Effective Continuum Legitimacy Thm.	[C]	4.2, 4.7	Book VII; all branch books using calc./PDE language

Scholium 9.1 ([R]). Book VI is the thinnest book in the spine from a source-content perspective. Its theorems are mostly biconditionals: differential notation is legitimate iff it encodes the right thing; integral notation is legitimate iff it encodes the right thing. The conceptual weight lies in the definitions — what "encoding" means, what a certified observable family is, what an effective smooth representative is. Once those definitions are set correctly, the theorems follow directly. This thinness is not a weakness; it reflects that the book's job is tool-licensing, not physics. The physics

begins in the branch books, which must come to Book VI with their observable families in hand and earn the license.